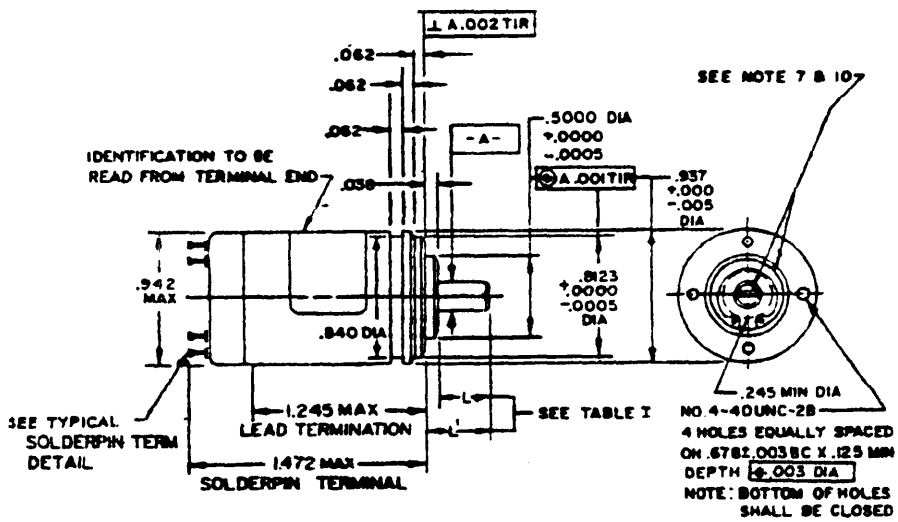
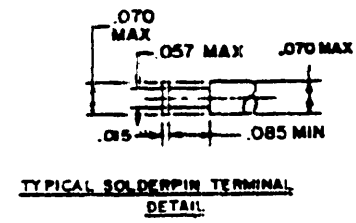
MILITARY SPECIFICATION SHEET

TRANSOLVER
TYPE 11.8 10R13Y440A

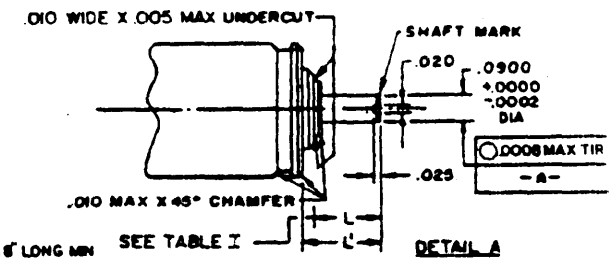
The complete requirements for procuring the transolver described herein shall consist of this document and the latest issue of MIL-T-83727.

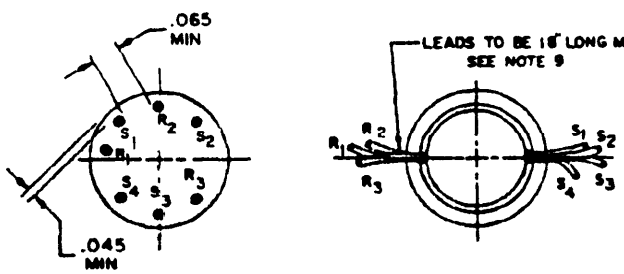
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





TYPICAL SOLDERPIN TERMINAL DETAIL





TYPICAL END TERMINATIONS SEE NOTE 4, 5 & 6

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 on decimals, ±2° on angles.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. Terminal lead wires shall depart through the back surface of the terminal end of the transolver in any acceptable manner.
5. Minor variations on unit configuration are permitted for undimensioned design detail.
6. Internal connections to terminals shall be welded or soldered with high temp solder SN-5 per QQ-S-571.
7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. transolver zero and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the transolvers to be furnished under this specification shall be subjected to the first article tests as specified in MIL-T-83727.
9. Leads shall be type E-28.
10. The front retainer shall be threaded or a snap ring.

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REQUIREMENTS:	VALUE	UNIT	TOLERANCE
Primary element	Rotor	-	-
Excitation frequency	400	Hz	±1%
Primary voltage	11.8	Volts	±1%
Primary current	92	Milliamperes	Max.
Primary power	.22	Watts	Nominal
D.C. resistance:			
Rotor	32	Ohms	±15%
Stator	175	Ohms	±15%
Impedance:			
Z_{ro}	34+j119	Ohms	R±15%+jx±10%
Z_{so}	234+j797	Ohms	R±15%+jx±10%
Z_{ss}	330+j95	Ohms	Referenced
Transformation ratio:			
Output/input	2.47	-	±2%
Phase shift	11.5	Degrees	±1.5°
Electrical error	7.5	Minutes	Max.
Interaxis error	5	Minutes	Max.
Null voltage:			
Fundamental	30	MV rms	Max.
Total	40	MV rms	Max.
Friction torque:			
At 23° C	.04 (3)	Oz-in. (Gm-Cm)	Max.
At -55° C and +125° C	.1 (9)	Oz-in. (Gm-Cm)	Max.
Radial play	.0004 (Using 4 oz. load)	Inches	Max.
End play	.0001-.0008 (Using 8 oz. load)	Inches	Min. - Max.
Temperature rise	5	°C	Max.

Part number: M83727/9 and a dash number from table I.

TABLE I. Dash number and characteristics.

Dash no.	Shaft diameter	Shaft style	Termination	Shaft length	
				L ref	L' dim to first mounting surface (±.010)
-001	.0900	Plain	Wire leads	.255	.293
-002	.0900	Plain	Solder pins	.255	.293
-003	.0900	Plain	Wire leads	.337	.375
-004	.0900	Plain	Solder pins	.337	.375
-005	.0900	Plain	Wire leads	.380	.418
-006	.0900	Plain	Solder pins	.380	.418
-007	.0900	Plain	Wire leads	.437	.475
-008	.0900	Plain	Solder pins	.437	.475

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Custodians:
Army - MU
Navy - AS
Air Force - 80

Review activities:
Army - EL, AV
Navy - SH, EC
Air Force - 11, 17
DSA - ES

User activities:
Navy - CG, MC
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0265)

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